CSS123P | BM17

Programmed by Group 5

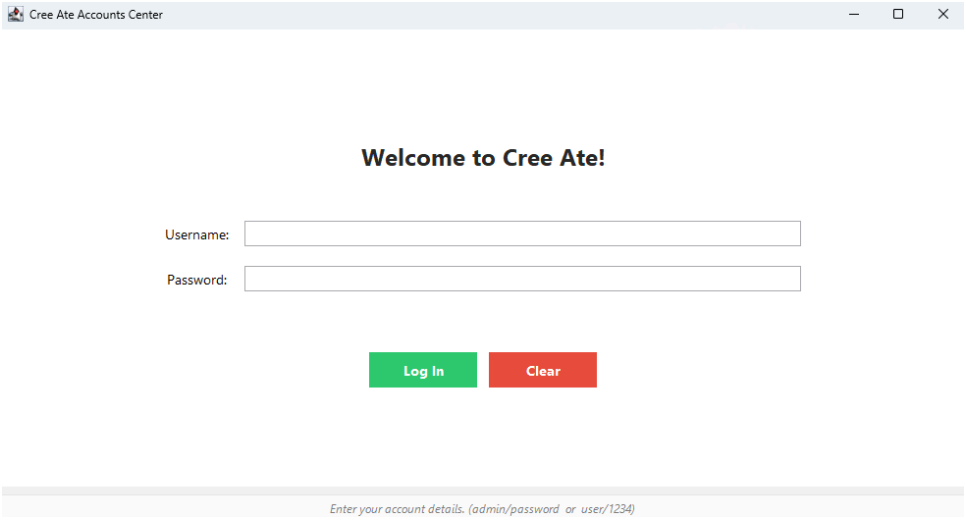
Laboratory Exercise 6

Members of the WindowB Mini Project	
Names	Roles
Joseph Gabriel A. Roces	Quality Engineer
Sean Patrick Brix V. Salamera	Data Engineer
Marvin Karl R. Sangco	SWE - BackEnd
Leonard Vincent L. Camat	SWE - FrontEnd

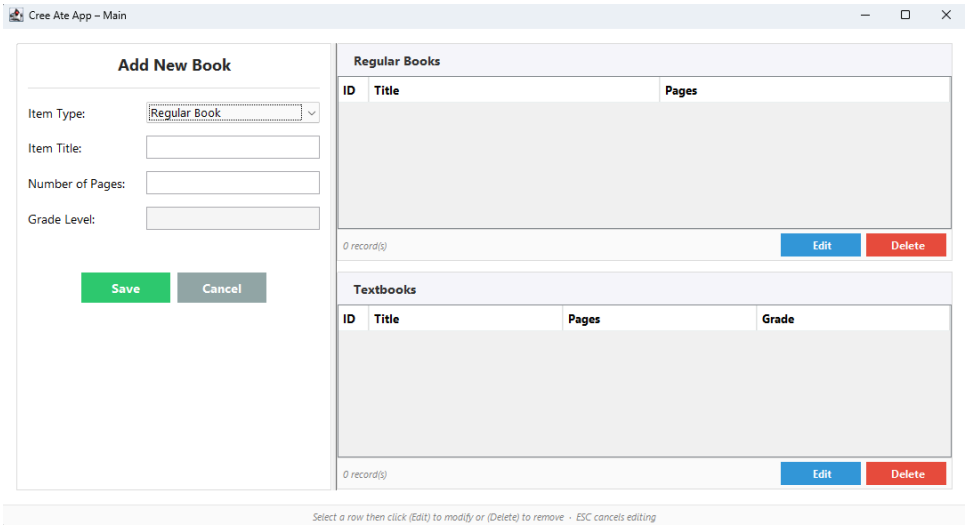
All the workflow details are in the GitHub repository and in the *README.md* file. The workflow may have a few changes and may not be accurate from the images below, but the main idea stays the same, mostly UI optimization changes.
GitHub Repository Link:
https://github.com/JGR-Dev-Scholar/Mini-Library-Management

Test Cases for LoginGUI

==========[Login Test Case

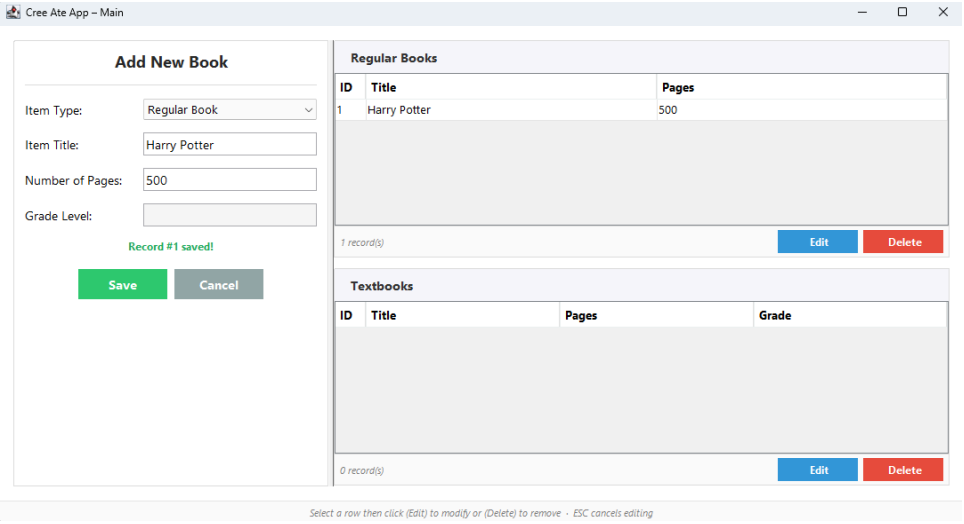


==========[MainGUI Test Case

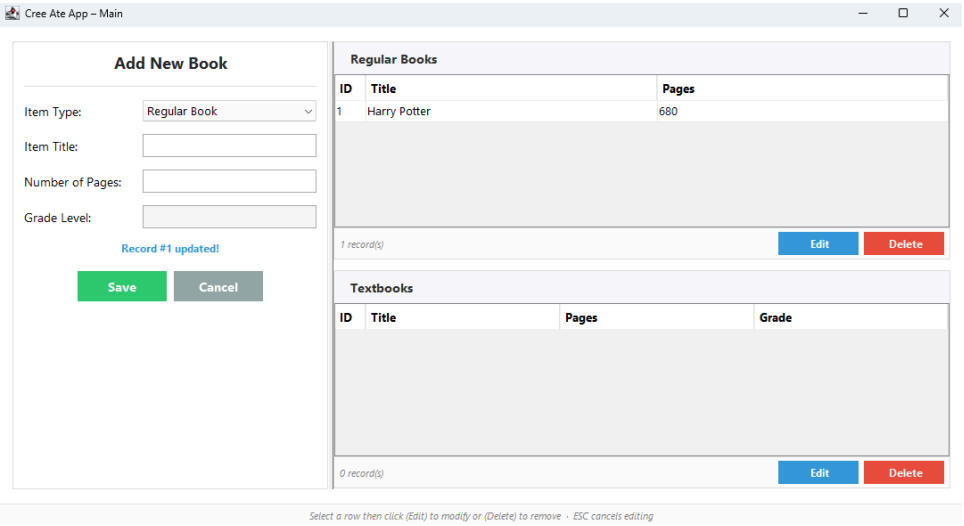


Test Cases for CRUD Functionality

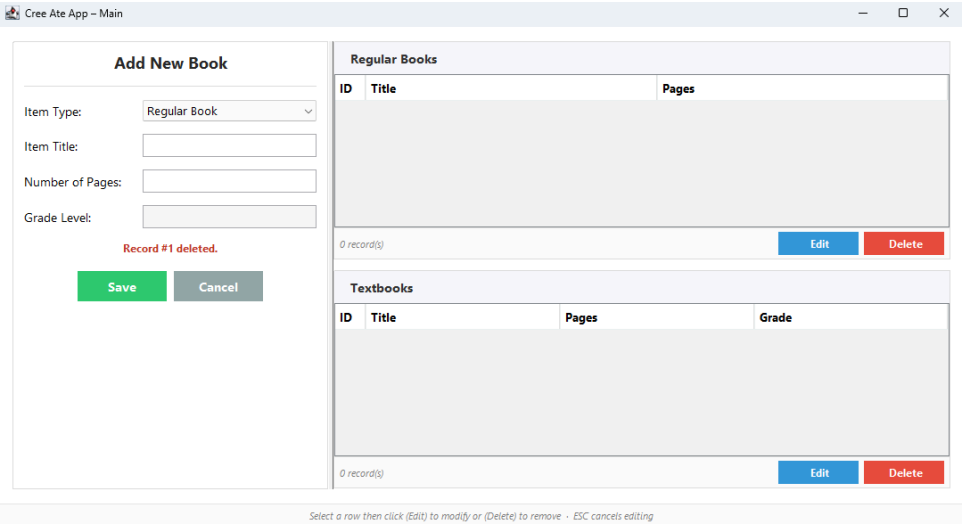
==========[Create for Regular Book



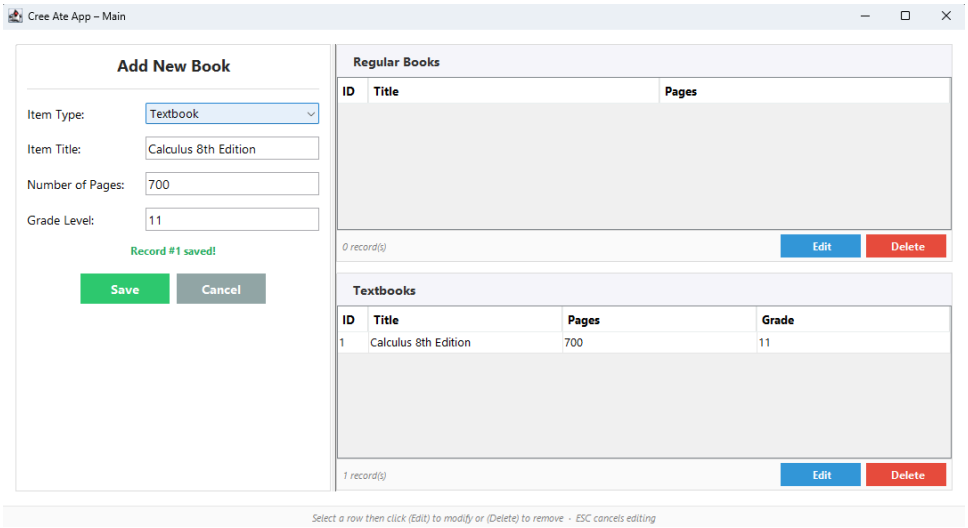
==========[Read & Update Regular Book



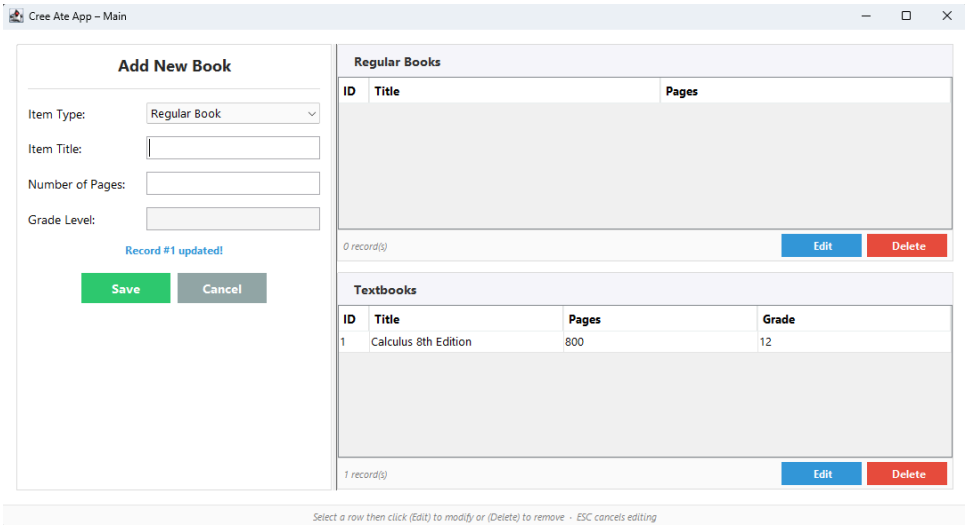
==========[Delete Regular Book



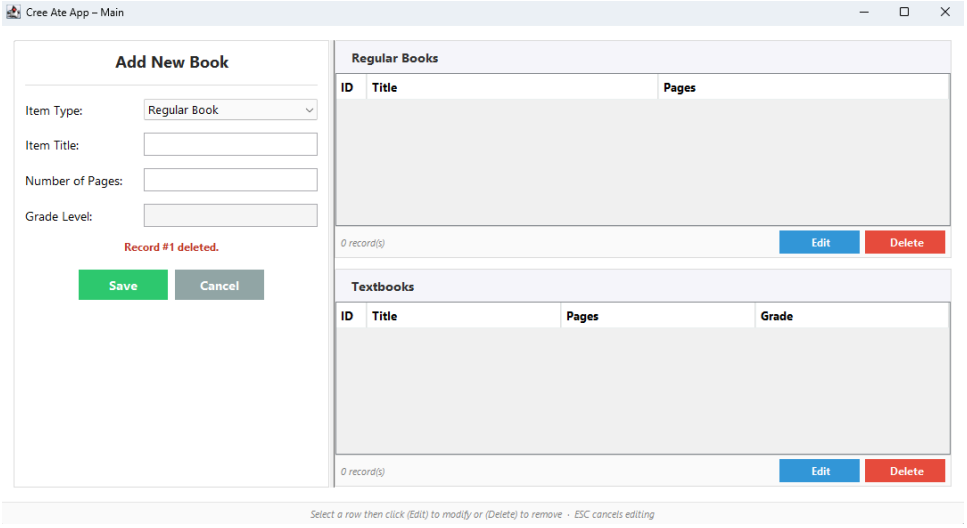
==========[Create for TextBook



==========[Read & Update TextBook

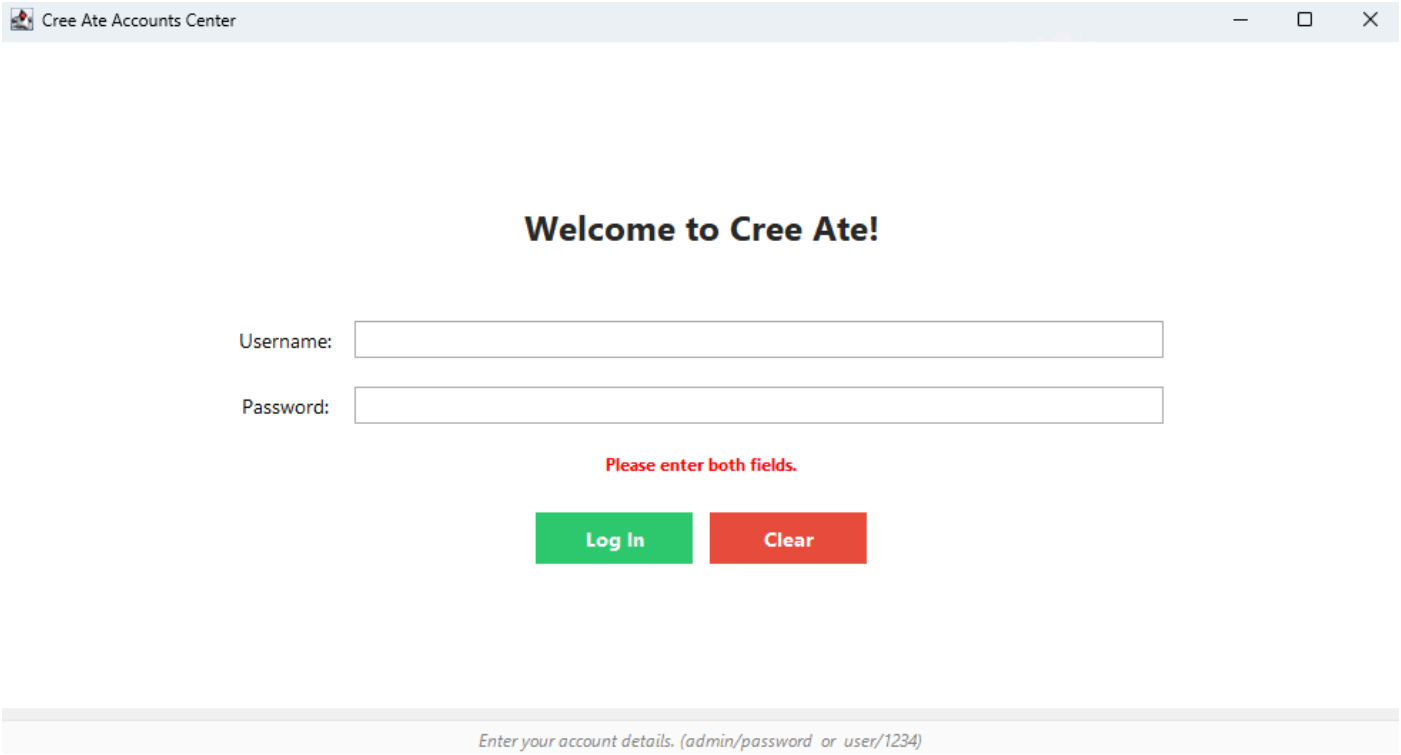


==========[Delete TextBook

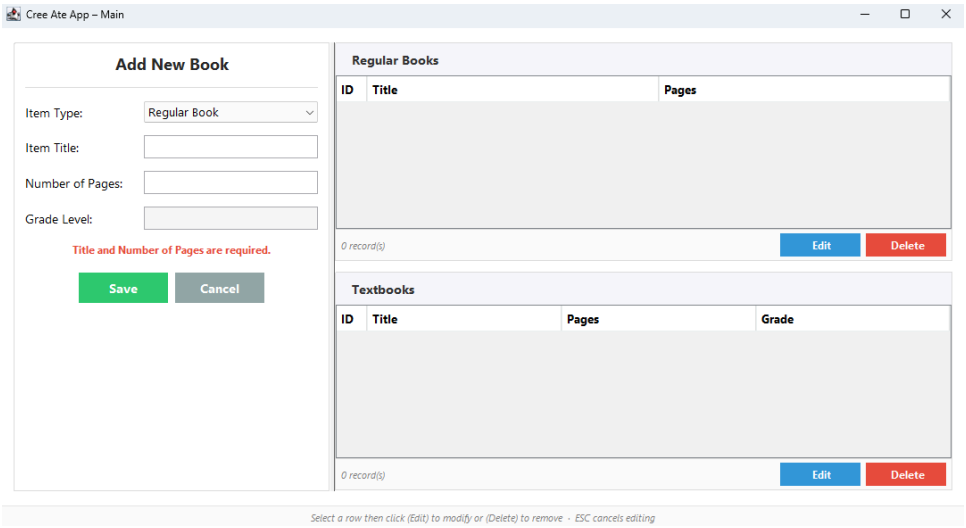


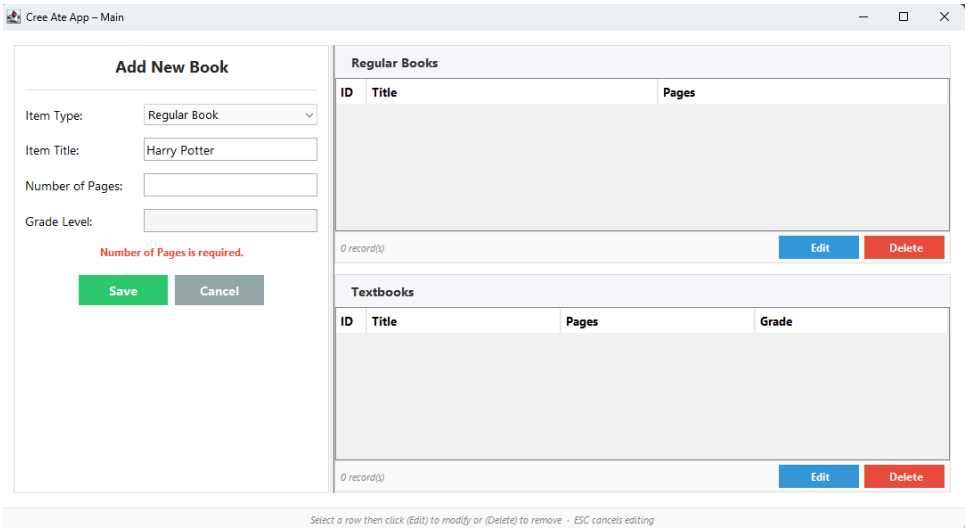
Test Cases for Error Handling/Exceptions

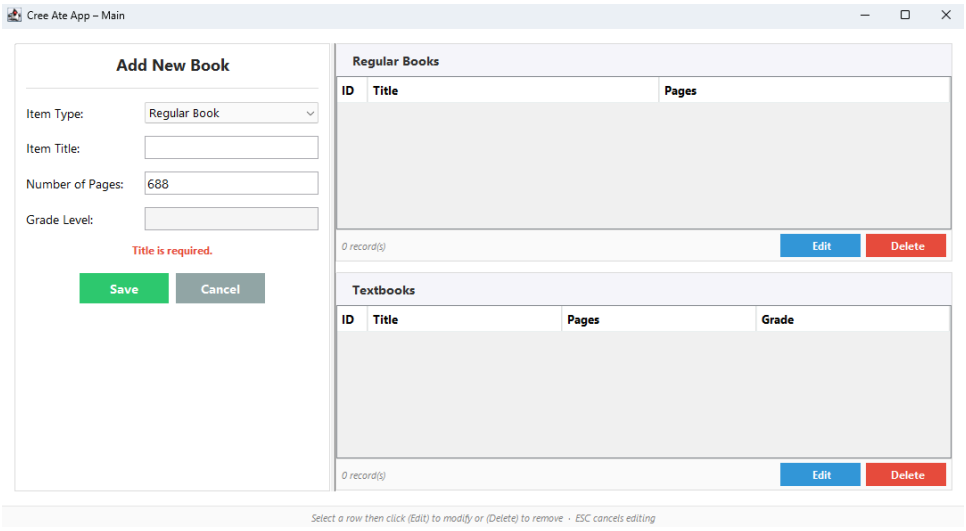
==========[LoginGUI Errors

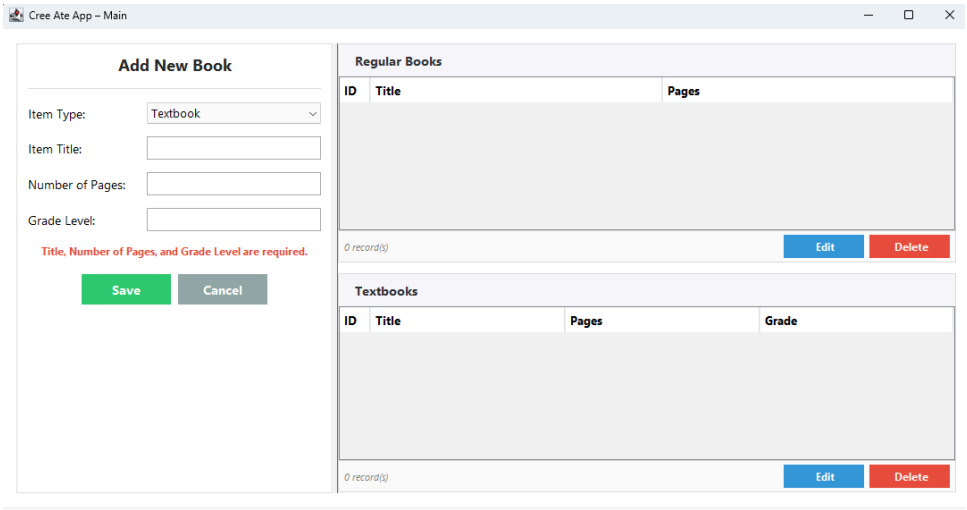


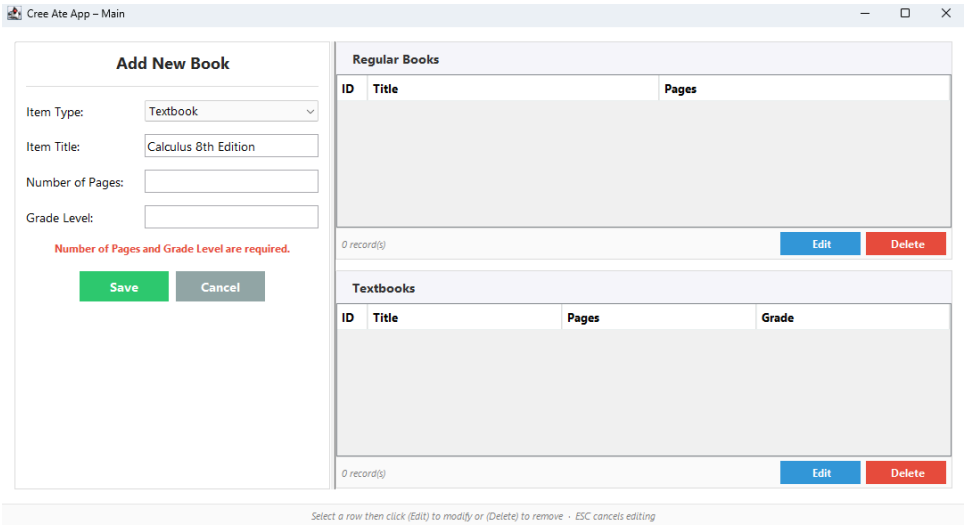
==========[General Errors

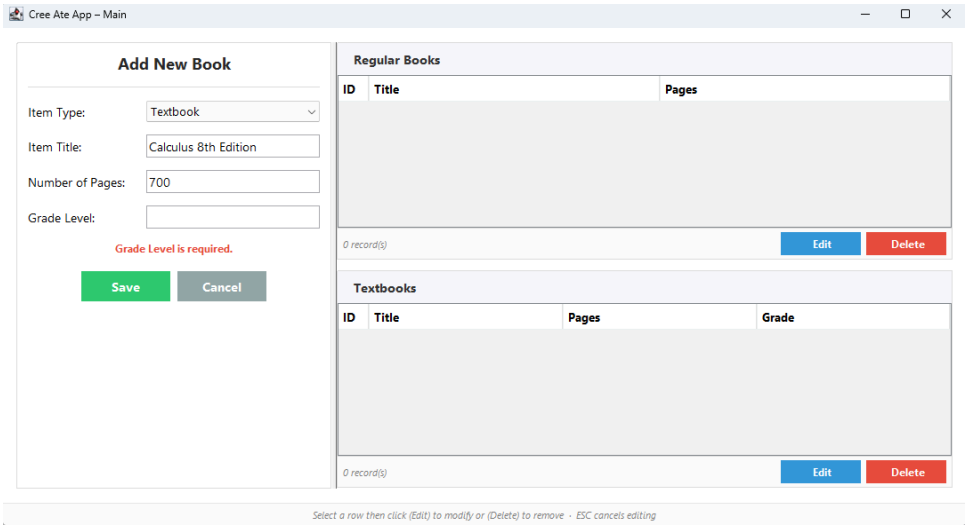




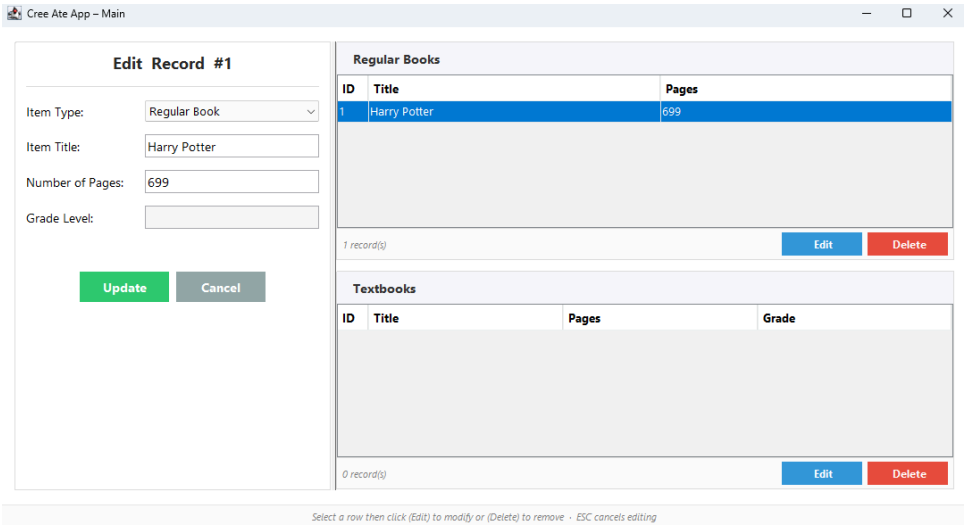

Cree Ate App – Main
Add New Book
Item Type:
Regular Book
Item Title:
Number of Pages:
688
Grade Level:
Title is required.
Save
Cancel
Regular Books
ID
Title
Pages
0 record(s)
Edit
Delete
Textbooks
ID
Title
Pages
Grade
0 record(s)
Edit
Delete
Select a row then click (Edit) to modify or (Delete) to remove · ESC cancels editing

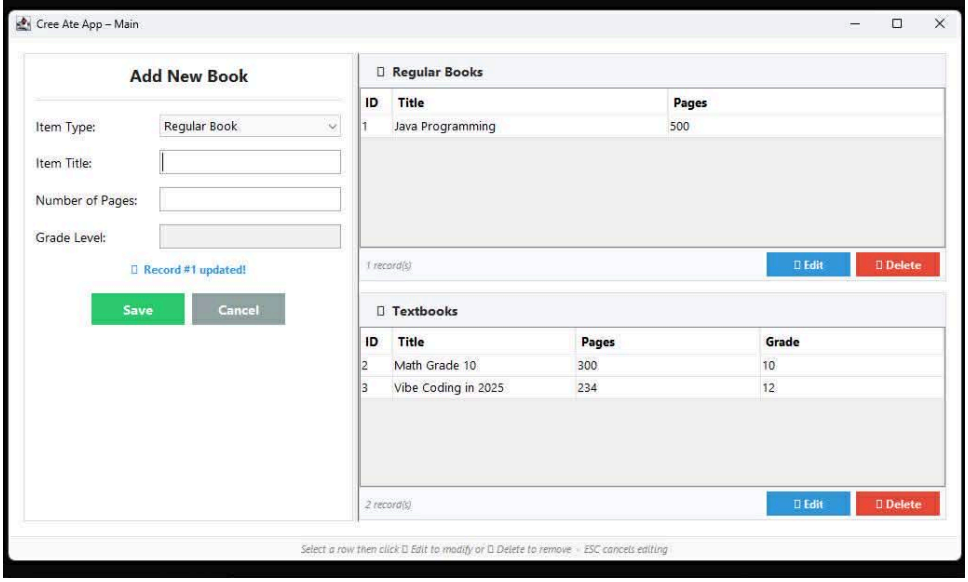

Cree Ate App – Main
Add New Book
Item Type:
Textbook
Item Title:
Number of Pages:
Grade Level:
Title, Number of Pages, and Grade Level are required.
Save
Cancel
Regular Books
ID
Title
Pages
0 record(s)
Edit
Delete
Textbooks
ID
Title
Pages
Grade
0 record(s)
Edit
Delete
Select a row then click (Edit) to modify or (Delete) to remove · ESC cancels editing


Cree Ate App – Main
Add New Book
Item Type:
Textbook
Item Title:
Calculus 8th Edition
Number of Pages:
Grade Level:
Number of Pages and Grade Level are required.
Save
Cancel
Regular Books
ID
Title
Pages
0 record(s)
Edit
Delete
Textbooks
ID
Title
Pages
Grade
0 record(s)
Edit
Delete
Select a row then click (Edit) to modify or (Delete) to remove · ESC cancels editing



Additional for Edit or Update



DataBase Connectivity

```
MySQL 8.0 Command Line Cli
| pages       | int          | NO   |     | NULL    |                |
| book_type   | varchar(20)  | NO   |     | NULL    |                |
| grade_level | int          | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> DESCRIBE BOOKS;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| id          | int          | NO   | PRI | NULL    | auto_increment |
| title       | varchar(200) | NO   |     | NULL    |                |
| pages       | int          | NO   |     | NULL    |                |
| book_type   | varchar(20)  | NO   |     | NULL    |                |
| grade_level | int          | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> SELECT title FROM books;
+---------------------+
| title               |
+---------------------+
| Java Programming    |
| Math Grade 10       |
| Vibe Coding in 2025 |
+---------------------+
3 rows in set (0.00 sec)

mysql> SELECT
    -> title
    -> ,id
    -> ,pages
    -> ,book_type
    -> ,grade_level
    ->
    -> FROM
    -> books
    -> ;
+---------------------+----+-------+--------------+-------------+
| title               | id | pages | book_type    | grade_level |
+---------------------+----+-------+--------------+-------------+
| Java Programming    |  1 |   500 | Regular Book |        NULL |
| Math Grade 10       |  2 |   300 | Textbook     |          10 |
| Vibe Coding in 2025 |  3 |   234 | Textbook     |          12 |
+---------------------+----+-------+--------------+-------------+
3 rows in set (0.00 sec)

mysql>
```